MZUZU UNIVERSITY

Department of Mathematics

EMT 1201 - Calculus

Supplementary/Deferred Examinations

20 September, 2016 Time: 3 hours

1. Check that the paper has 11 questions.
2. Attempt all questions
3. Show all your working
4. Write answers in the space provided
5. Use proper notation
6. Simplify your answers as far as possible
7. **No Calculators**

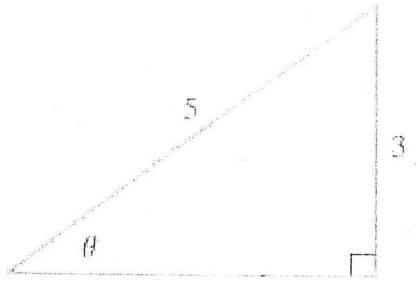
Name _____

Reg Number , Programme: _____

Lecturer's Name: _____

Question:	1	2	3	4	5	6	7	8	9	10	11	Total
Points:	10	8	9	6	5	7	9	6	7	6	7	80
Score:												

1. (a) Let θ be the acute angle defined by the figure below. Show that $5\sin(x+\theta) = 4\sin x + 3\cos x$. (5)



(b) Compute $\sin(\alpha - \beta)$ given that $\sin\alpha = \frac{12}{13}$ where $\frac{\pi}{2} < \alpha < \pi$ and $\cos\beta = -\frac{3}{5}$ where $\pi < \beta < \frac{3\pi}{2}$.

2. If $a\sin^2\alpha + b\cos^2\alpha = 1$, show that

(a) $\sin^2\alpha = \dfrac{1-b}{a-b}$ (4)

(b) $\tan\alpha = \dfrac{b-1}{1-a}$ (4)

3. (a) Evaluate each expression

i. $\cos(-585°)$ (2)

ii. $\sin 405°$ (2)

iii. $\cos\left(\frac{17\pi}{6}\right)$ (2)

(b) Use a product to sum formula to simplify $\cos\frac{5\pi}{12}\cos\frac{\pi}{12}$ (3)

4. (a) Evaluate $\lim_{t\to 9}\frac{9-t}{3-\sqrt{t}}$. (3)

(b) Find $\lim_{x\to 2^+}\frac{|x-2|}{x-2}$

5. Find the horizontal and vertical asymptotes of the curve $y = \dfrac{e^x}{e^x - 5}$. (5)

6. Consider the function

$$f(x) = \begin{cases} x^2 & \text{if } x \le 2 \\ mx + b & \text{if } x > 2 \end{cases}$$

where m and b are constants

(a) Find a relation between m and b that makes f is *continuous* at $x = 2$. (3)

(b) Determine what values of m, b make the function *differentiable* at $x = 2$ (4)

[Hint: what do the two sides want the slope to be at 2?]

7. (a) Differentiate the following functions

i. $f(t) = \sqrt{t}(a + bt)$ (3)

ii. $y = \frac{t^2+2}{t^4+3t^2+1}$ (3)

(b) Find the second derivative of the function $y = xe^{cx}$ (3)

8. Find derivative of $g(x) = \sqrt{1+2x}$ by evaluating the limit $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$. (6)

9. (a) Show that $\frac{d}{dx}(\sec x) = \sec x \tan x$ (3)

(b) If $f(x) = \sec x$, find $f'\left(\frac{\pi}{4}\right)$ (4)

10. Suppose $f(5) = 1$, $f'(5) = 6$, $g(5) = -3$ and $g'(5) = 2$. Find

(a) $(fg)'(5)$ (3)

(b) $\left(\frac{f}{g}\right)'(5)$ (3)

11. Find the absolute maximum and absolute minimum values of the function $f(x) = 2x^3 - 3x^2 - 12x + 1$ on the interval $[-2, 3]$. (7)